



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 21 – “Explained I”

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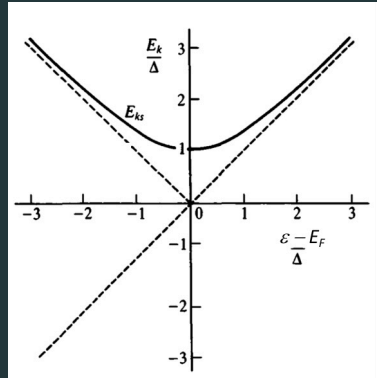
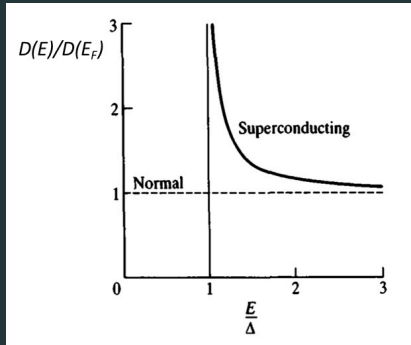
School of Physics & Astronomy

Previously, gap parameter Δ and T_c

$$\mathcal{H}_{\text{MF}} = \underbrace{\sum_{\mathbf{k}} \left((\epsilon_{\mathbf{k}} - \mu) - E_{\mathbf{k}} + \Delta \langle c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger \rangle \right)}_{\text{GS energy}} + \underbrace{\sum_{\mathbf{k}} E_{\mathbf{k}} \left(\gamma_{\mathbf{k}\uparrow}^\dagger \gamma_{\mathbf{k}\uparrow} + \gamma_{\mathbf{k}\downarrow}^\dagger \gamma_{\mathbf{k}\downarrow} \right)}_{\text{collection of new particles; bogoliubons}} \quad (1)$$

- $|BCS\rangle$ the vacuum of the γ particles
- $E_{\mathbf{k}}$ the excitation energy
- Δ the energy gap as $E_{\mathbf{k}} = \sqrt{(\epsilon_{\mathbf{k}} - \mu)^2 + |\Delta|^2}$

Density of States and Energy



Gap parameter Δ and T_c

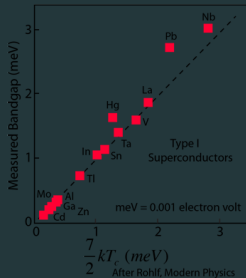
$$k_B T_c = 1.13 \hbar \omega_D e^{-1/D(E_F)|V|} \quad (2)$$

Comparing with

$$\Delta(0) = 2 \hbar \omega_D e^{-1/D(E_F)V} \quad (3)$$

gives

$$2\Delta(0) = 3.52 k_B T_c. \quad (4)$$



Finite temperature

Changing the sum

$$\frac{1}{V} = \frac{1}{2} \sum_k \frac{1}{E_k} \tanh \frac{E_k}{2k_B T}, \quad (5)$$

to an integral form, one could obtain for $T \approx T_c$

$$\frac{\Delta(T)}{\Delta(0)} \approx 1.74 \left(1 - \frac{T}{T_c}\right)^{1/2} \quad (6)$$

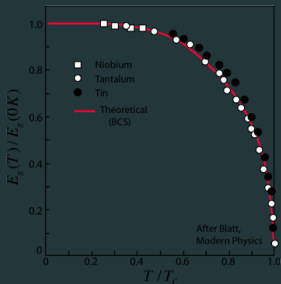


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Tunneling current

Tunneling current

Generally, from metal 1 to metal 2 (or superconductors)

$$I(V)_{1 \rightarrow 2} \propto \int \underbrace{D_1(E)f(E)}_{\text{\# of occupied states}} \underbrace{D_2(E + eV) [1 - f(E + eV)]}_{\text{\# of empty states}} dE, \quad (7)$$

$$I(V)_{1 \leftarrow 2} \propto \int D_2(E + eV)f(E + eV)D_1(E) [1 - f(E)] dE, \quad (8)$$

$$I(V) = I(V)_{1 \rightarrow 2} - I(V)_{1 \leftarrow 2} \quad (9)$$

$$\propto \int D_1(E)D_2(E + eV) [f(E) - f(E + eV)] dE \quad (10)$$

Normal to normal

From normal to normal metals,

$$\begin{aligned} I_{nn} &\propto \int D_1(E) D_2(E + eV) [f(E) - f(E + eV)] dE \\ &\propto D_1(E_F) D_2(E_F) \int [f(E) - f(E + eV)] dE \\ &\propto D_1(E_F) D_2(E_F) eV \\ &\propto V \end{aligned}$$

Ohmic behaviour ($I \propto V$).

Normal to superconductor

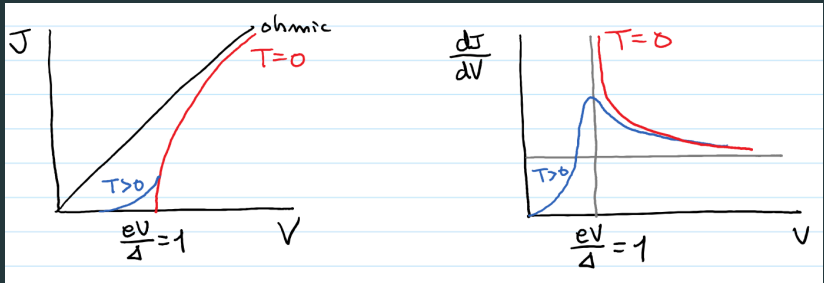


Figure 1: Current-voltage characteristic (left) and differential conductance versus voltage (right).

Conductance ($T = 0$)

$$G_{ns} = \frac{dI}{dV} = G_{nn} \frac{D_s(eV)}{D_n(E_F)} \quad (11)$$

Heat capacity

Specific heat

Entropy for fermions (e.g., electrons or γ particles)

$$S = -2k_B \sum_k [(1 - f_k) \ln(1 - f_k) + f_k \ln f_k]$$

where $f_k = (1 + e^{E_k/k_B T})^{-1}$ and $E_k = \sqrt{(\epsilon_k - \mu)^2 + \Delta^2}$

Electronic specific heat

$$C_s = T \frac{\partial S}{\partial T} \quad (12)$$

$$= -2k_B T \sum_k \ln \left(\frac{f_k}{1 - f_k} \right) \frac{\partial f_k}{\partial T} \quad (13)$$

Specific heat jump (1/2)

Electronic specific heat for normal state

$$C_n = \gamma T = \frac{2\pi^2}{e} D(E_F) k_B^2 T \quad (14)$$

Then at T_c

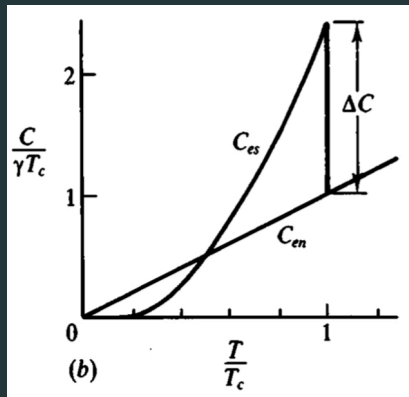
$$C_s - C_n = D(E_F) \left(\frac{-d\Delta^2}{dT} \right) \Big|_{T_c}, \quad (15)$$

which leads to

$$\lim_{T \rightarrow T_c} \frac{C_s(T) - C_n(T)}{C_n(T)} = 1.43 \quad (16)$$

Specific heat jump (2/2)

$$\lim_{T \rightarrow T_c} \frac{C_s(T) - C_n(T)}{C_n(T)} = 1.43 \quad (17)$$



Experimental specific heat

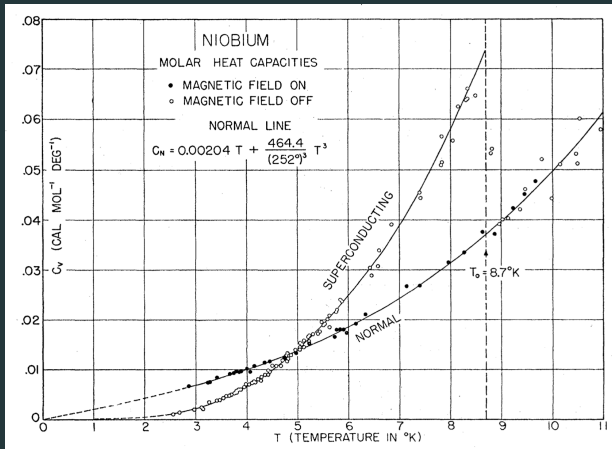


Figure 2: Heat capacity of Niobium in normal and sueprconducting states.
(from A. Brown et al., Phys. Rev. 92, 52 (1953)).